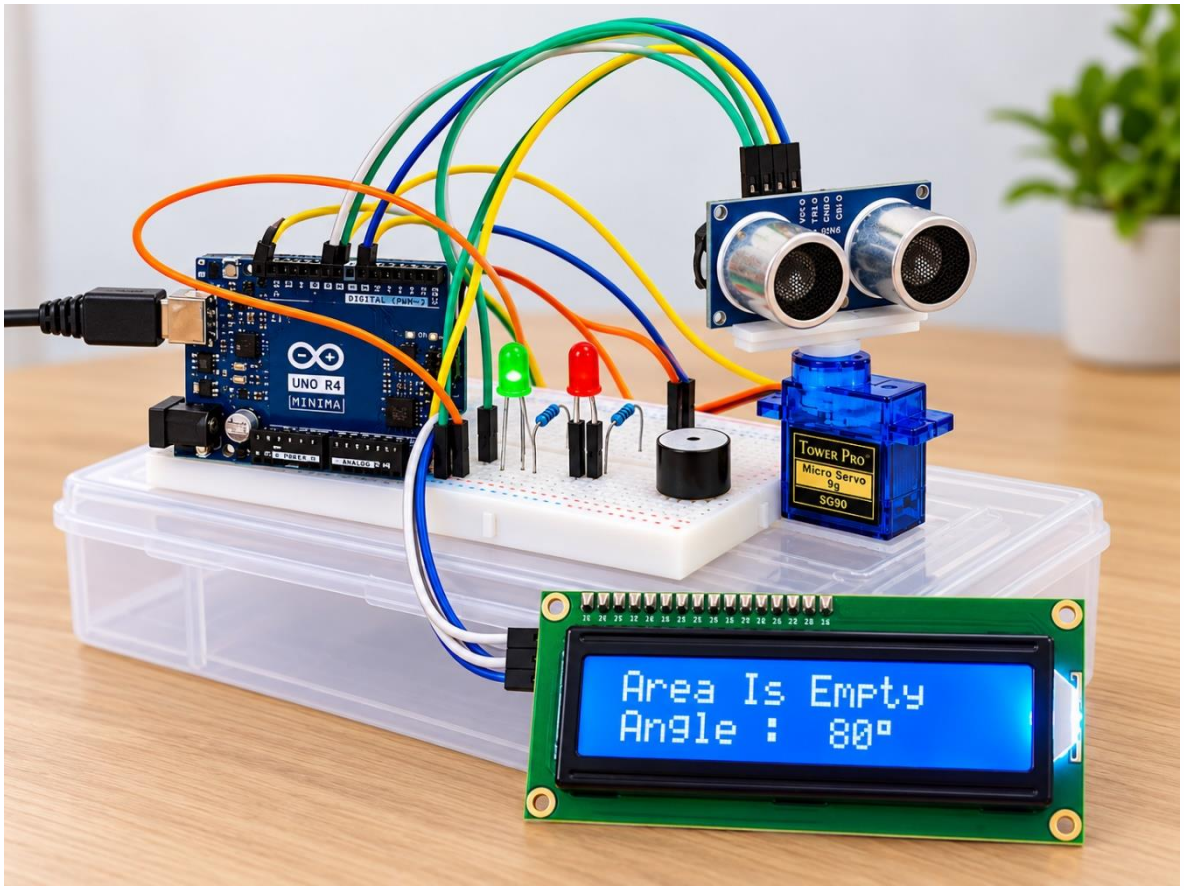
LAUNCH THE PROJECT - 01

Smart Radar Scanning System

(Advance level Project)

On AYNOP® UNO Launchpad Kit



Author: AYNOP Enterprises | Doc Version: 1.0 | Date: 01-09-2025

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1. Overview

The **Smart Radar Scanning System** is an advanced project that combines a servo motor, ultrasonic sensor, LEDs, buzzer, and 16x2 I2C LCD to create a simple radar-style object detection system.

The SG90 servo continuously scans from **0° to 180° and back**, while the HC-SR04 ultrasonic sensor checks for nearby objects.

The system provides clear visual and audio feedback:

- **Green LED ON** → Area is clear.
- **Red LED ON** → Object detected**.**
- **Buzzer ON** → Object detected.
- **LCD** → Displays the current radar status.

The system also uses consecutive sensor readings to reduce false detections caused by occasional unstable ultrasonic readings.

Through this project, you will learn how multiple components can work together as a complete sensing and alert system.

2. Components Required

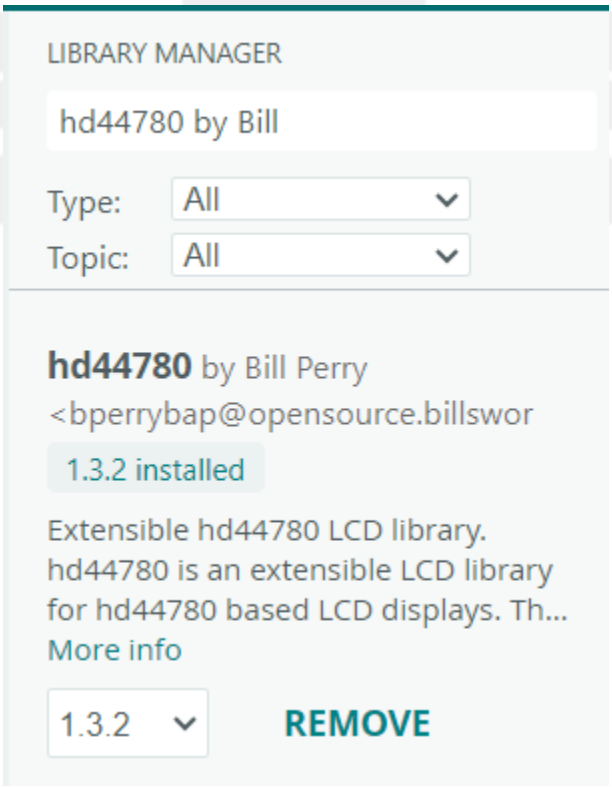
- Arduino® UNO R4 Minima board
- USB Type-C data cable
- HC-SR04 Ultrasonic Sensor
- TowerPro SG90 Micro Servo
- 16x2 I2C LCD Display
- 1 × 5mm Red LED
- 1 × 5mm Green LED
- 2 × 220Ω resistors
- 5V Active Buzzer
- 100uf Capacitor
- Jumper wires
- Breadboard
- Double Sided Tap (⚠ **Not included** in this kit – must be purchased separately to stick the SG90 motor and ultrasonics sensor HCS-SR04)

3. Software Required

- Arduino IDE (v2.3.7 or later)
- Required library – 01: **hd44780** for 16x2 I2C LCD Display
- Required library – 02: **Servo (built-in with Arduino IDE)**
- Required library – 03: **Wire/I2C (built-in with Arduino IDE)**

Installing the Library:

- Open Arduino IDE.
- Go to **Sketch → Include Library → Manage Libraries**
- Search for **hd44780 by Bill.**
- If a required library is already installed, **skip its installation**.
- Install the library.



Note:
We assume the **Arduino UNO R4 Minima board package** is already installed on your machine, as explained in the 00_Getting_Started/00_GettingStarted_Arduino_R4_Minima guide. If it is not installed, please refer to that document and complete the installation before proceeding.

4. Hardware Setup

This section explains how to connect the sensors and LCD to build the Smart Radar Scanning System.

4.1 Wiring Diagram

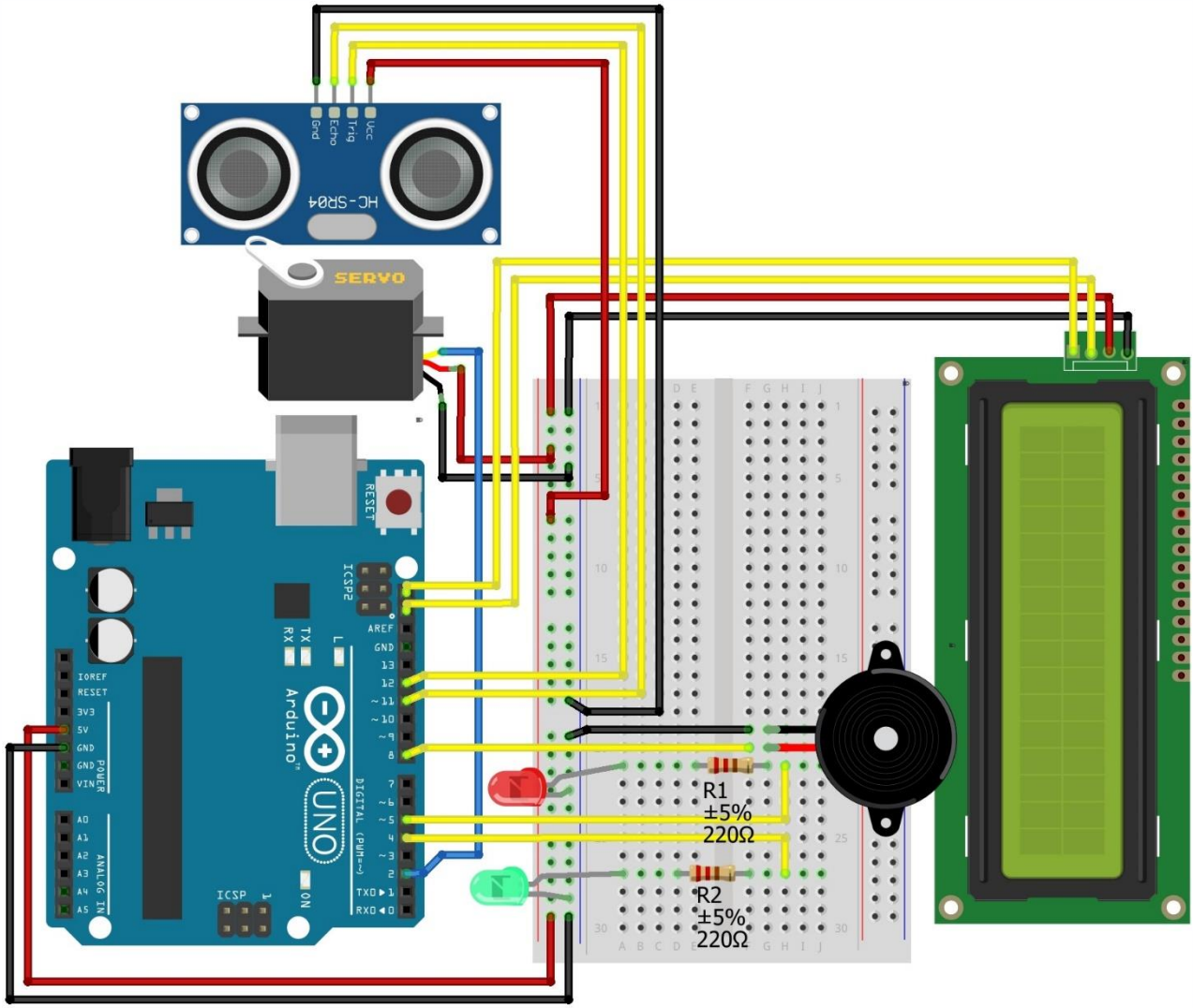


Figure 4.1 – Wiring diagram for Smart Radar Scanning System

Tip: I2C uses the same two wires (SDA, SCL) for multiple devices. Each device has a unique **I2C address**.

5. Principle – How It Works

The Smart Radar Scanning System works by combining a **servo motor and ultrasonic sensor** to scan the surrounding area and detect nearby objects.

The **SG90 servo** moves the ultrasonic sensor continuously from **0° to 180° and back**, allowing the sensor to check different directions.

At each position, the **HC-SR04 ultrasonic sensor** sends an ultrasonic pulse and measures the time taken for the echo to return. The Arduino uses this time to calculate the approximate distance of an object.

Object Detection

The system considers an object detected when it is within the configured **20 cm detection range**.

To make the detection more reliable, the system checks the sensor reading multiple times before changing the confirmed status. This helps reduce false alerts caused by occasional unstable readings.

System Response:

When the area is clear:

- 🟢 Green LED → ON
- 🔴 Red LED → OFF
- Buzzer → OFF
- LCD → Area Is Empty
- LCD shows the current scanning angle

When an object is detected:

- 🔴 Red LED → ON
- 🟢 Green LED → OFF
- Buzzer → ON
- LCD → OBJECT DETECTED!
- LCD shows the measured distance

This project demonstrates how a **sensor, motor, display, LEDs, and buzzer can work together as one complete detection system**, similar to the basic idea of a radar scanner.

6. Procedure – Steps to Run

1. **Build the Circuit**
   a. Connect the servo, ultrasonic sensor, LEDs, buzzer, and LCD according to the wiring diagram (Figure 4.1).

2. **Connect the Board**
   a. Use a USB Type-C data cable to connect your UNO R4 Minima to your computer.

3. **Open the Project Code**
   a. Simply **double-click** the file *01_Smart_Radar_Scanning_System.ino* in the project folder, and it will open directly in the Arduino IDE (if installed).

4. **Confirm Board Selection**
   a. The IDE usually auto-detects the UNO R4 Minima if the package is installed.
   b. If not installed, refer to the *00_Getting_Started/00_GettingStarted_Arduino_R4_Minima* document to install the necessary board package.
   c. Verify that *Arduino UNO R4 Minima* is selected in the IDE's board selector (top toolbar).

5. **Upload the Code**
   a. Click the **Upload** button (arrow icon) in the top-left corner of the IDE.
   b. Wait until the console displays "**Done uploading**."

6. **Observe the Behaviour**
   a. The servo continuously scans from 0° to 180° and back.
   b. When no object is detected within the 20cm detection range, the green LED remains ON and the LCD displays "Area Is Empty" and show the current scanning angle on LCD.
   c. When an object is detected within approximately 20 cm, the system confirms the detection and activates the red LED and buzzer and the LCD displays "OBJECT DETECTED!" along with the measured distance.

7. Expected Output

- **Start the Radar Scan:**
  - Power ON the Arduino and allow the system to start scanning.
  - Watch the SG90 servo sweep from 0° to 180° and back, just like a simple radar scanner.
  - When the area is clear:
    - 🟢 Green LED remains ON.
    - 🔴 Red LED remains OFF.
    - Buzzer remains OFF.
    - LCD displays “Area Is Empty”.

- **Bring an Object into the Scan Area:**
  - Slowly move your **hand or another object** into the sensor's scanning area.
  - Keep the object within approximately **20 cm** of the ultrasonic sensor.
  - As the sensor points toward the object, the system will detect it.
  - The **red LED turns ON** and the **buzzer sounds**.
  - The LCD displays **“OBJECT DETECTED!”** along with the measured distance.

- **Move the Object Away:**
  - Slowly move your hand or object away from the sensor.
  - Observe the system as the sensor continues scanning.
  - Once the object is no longer detected, the system returns to the clear-area state:
    - 🟢 Green LED ON
    - 🔴 Red LED OFF
    - Buzzer OFF
    - LCD displays “Area Is Empty”

💡 **Fun Testing Tip: Play radar!** 🛠️ Move your hand slowly around the sensor and watch the servo scan. Try moving the object **toward and away from the sensor** and observe how the **servo, LEDs, buzzer, and LCD respond together**.

8. Code

The source code for this project is included in the downloaded folder:

📂 uno-launchpad-kit/03_Advanced_Projects/01_Smart_Radar_Scanning_System/
01_Smart_Radar_Scanning_System.ino

👉 **Tip:**

- To open the project, simply **double-click the .ino file**. If the Arduino IDE is installed, it will launch automatically and load the code.
- If you **haven't installed the Arduino IDE yet**, please refer to:
  📂 uno-launchpad-kit/00_Getting_Started/00_GettingStarted_Arduino_R4_Minima
  to **download and install it.**

8.1 Function References

- updateServoPosition() – Moves the servo gradually between 0° and 180° and reverses the scanning direction at the ends.
- measureObjectDistance() – Triggers the HC-SR04 ultrasonic sensor, measures the echo time, and calculates the approximate object distance in centimetres.
- updateDetectionStatus() – Checks consecutive distance readings to confirm whether an object is detected or the scanning area is clear.
- updateAlertIndicators() – Controls the red LED, green LED, and buzzer according to the confirmed detection status.
- displayRadarStatus() – Updates the 16×2 I2C LCD with the current radar status, scanning angle, or detected object distance.
- digitalWrite(pin, value) – Turns the connected LED or buzzer ON or OFF.
- pulseIn(pin, state, timeout) – Measures the duration of the ultrasonic echo signal returned by the HC-SR04 sensor.
- Servo.write(angle) – Moves the SG90 servo to the specified angular position.
- millis() – Returns the elapsed time since the Arduino started running and is used for timing without stopping the radar scanning process.

📚 **Library Reference:**

🔗 Wire Library — I2C communication library used internally by hd44780.
🔗 hd44780 — I2C LCD control library used to interface the 16×2 display.

📚 **For more details and advanced usage, visit:**
🔗 Arduino Language Reference — The official guide for all Arduino functions.

9. Troubleshooting Tips

- **Error while compiling (hd44780 library not found)?**
  - Open Arduino IDE → Library Manager.
  - Search for "hd44780 (by Bill Perry)" and install it.
  - Restart the IDE and compile again

- **LCD backlight ON but no text?**
  - Adjust the small **blue potentiometer** on the I2C adapter to set contrast.

- **LCD backlight not ON?**
  - I2C adapter boards have a small **jumper near the backlight pins**.
  - Ensure this jumper is connected; otherwise, the backlight will stay OFF

- **LCD shows nothing?**
  - Check wiring: VCC → 5V, GND → GND, SDA → D18, SCL → D19.
  - Verify the **I2C address** (0x27 or 0x3F)

- **Upload error in Arduino IDE?**
  - Verify that the correct board (**Arduino UNO R4 Minima**) is selected in the IDE.
  - Check that the correct **COM port** is chosen.

- **Servo not moving?**
  - Check that the servo is properly connected to the built-in servo connection.
  - Make sure the servo is not mechanically blocked.
  - Restart the Arduino and upload the code again
  - Alos verify that there are no Loose connections in the servo wire.

- **Ultrasonic sensor not detecting an object??**
  - Check the **TRIG** and **ECHO** connections as per fig 4.1.
  - Make sure the sensor is facing the object.
  - Place the object within the configured **20 cm detection range**.
  - Keep the sensor steady while testing

- **Red LED or Green LED not working?**
  - Check the LED connections and polarity.
  - Verify that the 220Ω resistors are connected correctly.
  - Make sure the LED wires are firmly connected to the breadboard.

- **Buzzer not sounding?**
  - Check the buzzer polarity and connection.
  - Make sure the active buzzer is connected correctly.
  - Verify that the buzzer connection is not loose.

- **Object detection appears unstable?**
  - Move the object slowly while testing.
  - Avoid placing the sensor too close to large surfaces that may reflect ultrasonic waves.
  - Keep the ultrasonic sensor firmly positioned.
  - The system uses consecutive readings to help reduce false detections.

- **Arduino resets when the servo moves?**
  - Check that all power and GND connections are secure.
  - Make sure there are no loose breadboard connections.
  - Avoid short circuits or incorrect power connections.

- **Still not working?**
  - Recheck all wiring connections carefully.
  - Disconnect power and rebuild the circuit step-by-step.
  - Test the system step-by-step, starting with the LCD, servo, and ultrasonic sensor
  - If still not work then Test each module (LCD, Servo, HCS-SR04) individually using basic projects

10. License

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11. Support & Feedback

We value your feedback and are happy to assist with any questions, troubleshooting, or suggestions you may have.

Email: *support@aynop.com*

When sending an email, please include your kit name (AYNOP® UNO Launchpad Kit) and, if applicable, the project name in the subject line. This will help our team respond faster and more accurately. We aim to respond to all queries within 2–3 business days. Your feedback helps us improve our products and create even better learning experiences for all Arduino beginners.